How Transformers Build Their Predictions

Layer-by-layer view into how transformers resolve their next token.

What is this?

At every layer of a transformer, we project the intermediate hidden state onto the model's vocabulary through the LM head and record the top predicted token along with its probability. This is done at **three points** within each layer:

- **Before attention** — the residual stream entering the layer
- **After attention** — the residual stream after the attention sublayer is added
- **After FFN** — the residual stream after the feed-forward sublayer is added

Reading the diagrams from bottom to top, you can observe how the model's "best guess" evolves as information flows through the network.

Two types of diagrams

Each example includes two visualizations:

Layer-by-layer flow (`*_flow.png`) — shows the top-1 token at each stage of each layer. Highlighted cells with ▲/▼ indicators mark where a sublayer meaningfully changed the prediction, making it easy to pinpoint exactly *where* and *how* the model arrives at its answer.

Token probabilities (`*_probs.png`) — a line chart tracking the probabilities of all tokens that were ever top-1 across layers. This reveals the full competition between candidates that the flow diagram compresses into a single winner per stage.

Examples

All examples use **Qwen/Qwen2.5-3B** with prompts in `Q: <question>\nA:` format. We trace the first generated token (the one right after `A:`).

Example	Prompt	Answer	Key pattern
Capital of Japan	What is the capital of Japan?	Tokyo	FFN at L30 pushes prob from 0.32 to 0.96 in one layer
Ulysses author	Who wrote Ulysses?	James	Literary tokens (writing, playwright) → James at L31
Closest planet		Mercury	

Example	Prompt	Answer	Key pattern
	Which planet is closest to the Sun?		Space tokens (planet, Earth, solar) → Mercury at L30
Lightest element	What is the lightest element?	Hydro	helium → hydrogen → "Hydro" tokenization switch at L35
Mona Lisa	Who painted the Mona Lisa?	Leonardo	Art tokens (painting, painter) → Leonardo at L31

Key observations

FFN as the knowledge retrieval mechanism

Across all examples, the critical moment — when the model switches from a wrong or vague token to the correct answer — is driven by an **FFN sublayer**. Attention may bring in related concepts (e.g., "Dublin" for Ulysses, "Earth" for the closest planet), but FFN is the one that commits to the factual answer: "Tokyo", "James", "Mercury", "hydrogen". This is consistent with the view that factual associations are encoded in feed-forward weights.

What different layers do

The 36 layers of the model divide into distinct functional zones:

L0–L10: Noise. The LM head projection at these depths is meaningless — it produces random multilingual fragments ("createElement", "cor", Arabic and Turkish tokens) at low probabilities. The hidden states have not yet aligned with the output vocabulary space.

L11–L14: Format recognition and domain detection. The model identifies the Q&A structure (tokens like "Answer", "Answers") and begins retrieving domain-relevant concepts — but not the specific answer yet. For chemistry questions, generic elements appear; for geography, generic locations.

L15–L27: Latent computation. The top-1 token often stalls on formatting tokens like "Answer" for many layers. On the surface nothing changes, but the residual stream is silently accumulating the information needed for the final answer.

L28–L33: Knowledge retrieval. The answer emerges and solidifies — this is where FFN sublayers fire the critical factual associations. Competing candidates may appear and get resolved within this window (e.g., helium vs hydrogen, Earth vs Jupiter vs Mercury).

L34–L35: Output formatting. The model shifts from *what to say* to *how to say it*. This includes tokenization decisions ("hydrogen" → "Hydro" + "gen" for capitalized output) and stylistic adjustments. The answer itself is already decided — these layers shape its surface form.

What is the capital of Japan?

Token flow

L0–L10 — noise. Random tokens cycle through the residual stream: "cor", "createElement", "//**". No semantic processing yet.

L11 — the model recognizes the Q&A format. "YES" (prob 0.27) appears briefly, then "Answers" (0.50) at L12. These are structural markers — the model understands it's in a question-answer context.

L13–L17 — turbulent zone. Top-1 token jumps between "Answers", "?", "Incorrect", and back. The model is processing the question structure but hasn't started retrieving factual knowledge.

L18–L24 — "Answer" locks in and holds for 7 layers. Prob fluctuates between 0.04 and 0.29. On the surface nothing changes, but the residual stream is accumulating geographic context from the question tokens ("capital", "Japan") through attention.

L25–L27 — subtle refinement: "Answer" → "(answer)". The parenthesized form suggests the model is transitioning from "I need to answer" to "I'm about to produce the answer". Prob is low (0.12–0.24) — the model hasn't committed yet.

L28–L29 — "(answer)" continues. Then at L29 output, FFN delivers the breakthrough: **"Tokyo"** appears as top-1 with prob 0.30. First appearance of the actual answer.

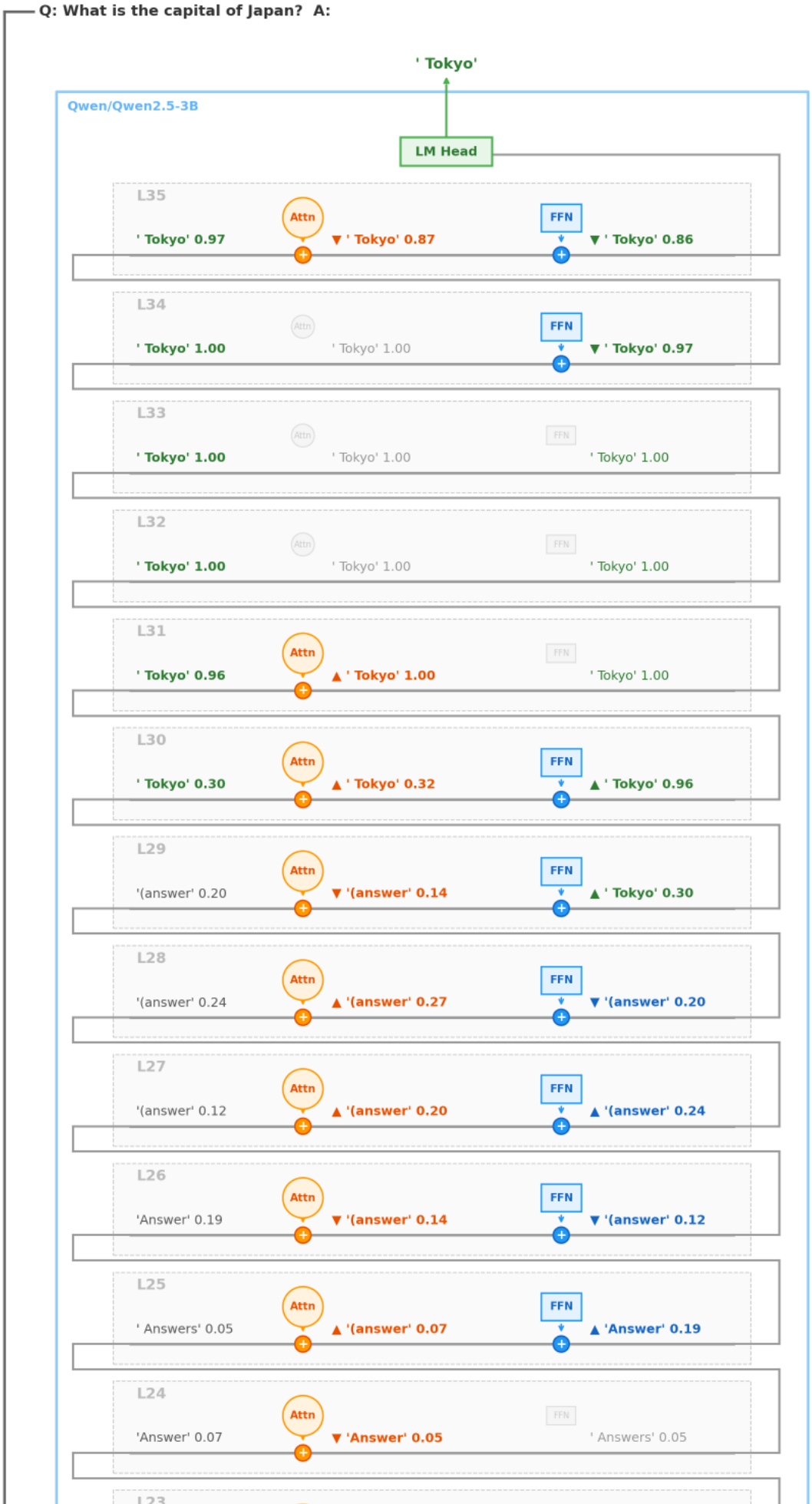
L30 — the critical layer. Attention slightly reinforces Tokyo (+0.02). Then FFN makes a massive push: prob jumps from 0.32 to **0.96** in a single layer. This is where the knowledge neurons fire — the FFN at L30 contains the "capital + Japan → Tokyo" association.

L31 — attention pushes Tokyo to **1.00**. Essentially certain. The answer is locked in.

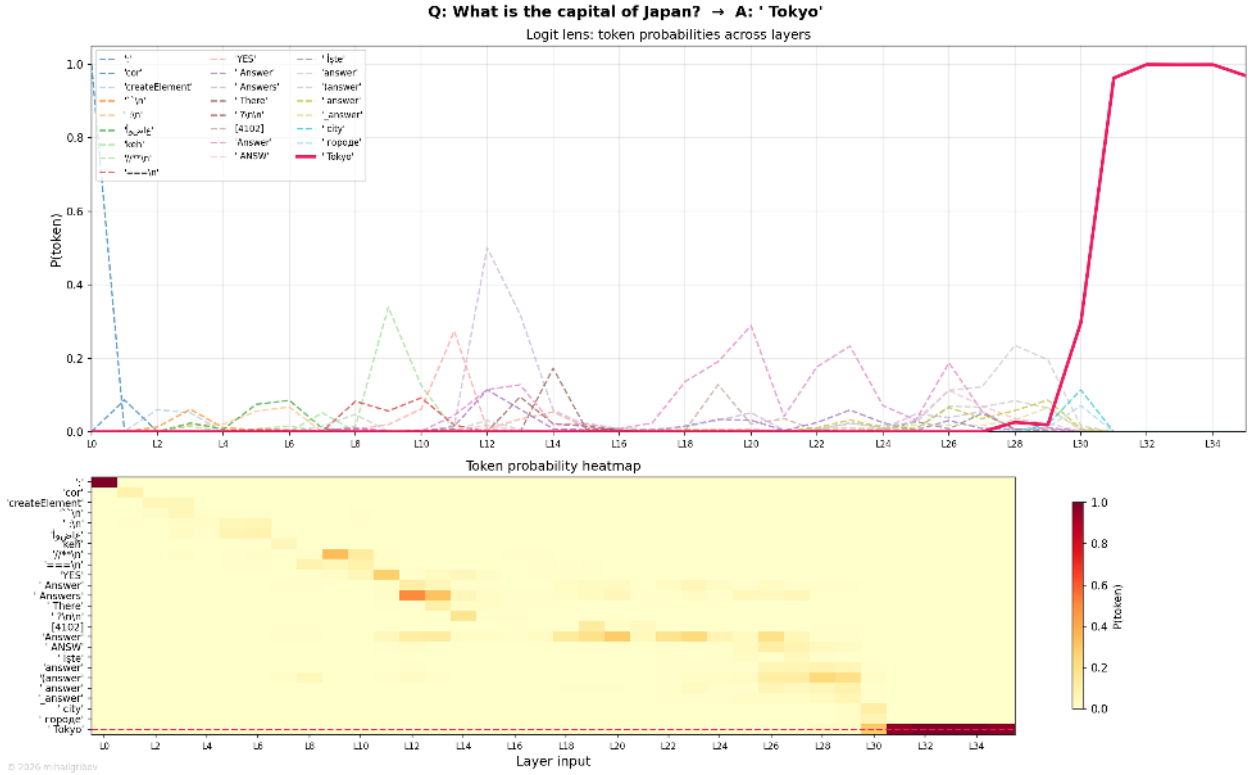
L32–L34 — maintenance. Both modules preserve Tokyo with prob 0.99–1.00. Minor FFN adjustments but nothing changes the outcome.

L35 — slight erosion. Attention drops prob to 0.87. This is normal for the last layer — the model is preparing for LM Head formatting.

Final output via LM Head: **Tokyo**.


Q: What is the capital of Japan? A:
' Tokyo'
Qwen/Qwen2.5-3B
LM Head
L35
' Tokyo' 0.97
Attn
▼ ' Tokyo' 0.87
FFN
▼ ' Tokyo' 0.86
L34
' Tokyo' 1.00
Attn
' Tokyo' 1.00
FFN
▼ ' Tokyo' 0.97
L33
' Tokyo' 1.00
Attn
' Tokyo' 1.00
FFN
' Tokyo' 1.00
L32
' Tokyo' 1.00
Attn
' Tokyo' 1.00
FFN
' Tokyo' 1.00
L31
' Tokyo' 0.96
Attn
▲ ' Tokyo' 1.00
FFN
' Tokyo' 1.00
L30
' Tokyo' 0.30
Attn
▲ ' Tokyo' 0.32
FFN
▲ ' Tokyo' 0.96
L29
'(answer' 0.20
Attn
▼ '(answer' 0.14
FFN
▲ ' Tokyo' 0.30
L28
'(answer' 0.24
Attn
▲ '(answer' 0.27
FFN
▼ '(answer' 0.20
L27
'(answer' 0.12
Attn
▲ '(answer' 0.20
FFN
▲ '(answer' 0.24
L26
'Answer' 0.19
Attn
▼ '(answer' 0.14
FFN
▼ '(answer' 0.12
L25
' Answers' 0.05
Attn
▲ '(answer' 0.07
FFN
▲ 'Answer' 0.19
L24
'Answer' 0.07
Attn
▼ 'Answer' 0.05
FFN
' Answers' 0.05
L23

Tokyo (red) is invisible for 29 layers, then a near-vertical rise from 0 to 1.0 in just 2 layers (L29–L31). "Answer" (green dashed) dominates the middle section but vanishes the moment Tokyo arrives. The heatmap shows the clean handoff between structural tokens and the factual answer.



Who wrote Ulysses?

Token flow

L0–L10 — noise. Standard early-layer behavior with random tokens.

L11–L12 — Q&A format recognized. "YES" → "Answers" (prob 0.50). The model knows it needs to produce an answer.

L13–L17 — literary domain activates. Interesting tokens appear: "writing", "playwright", "novelist". The model has narrowed down to "this is about an author" but hasn't picked a name. This is category-level retrieval — the right domain, no specific entity yet.

L18–L27 — "Answer" dominates for 10 layers. Prob varies (0.04–0.29). The residual stream is quietly accumulating information about "Ulysses" from the question through attention — building up the context that will trigger the right FFN neurons.

L28–L29 — transitional. "(answer)" appears, prob low. The model is close to committing.

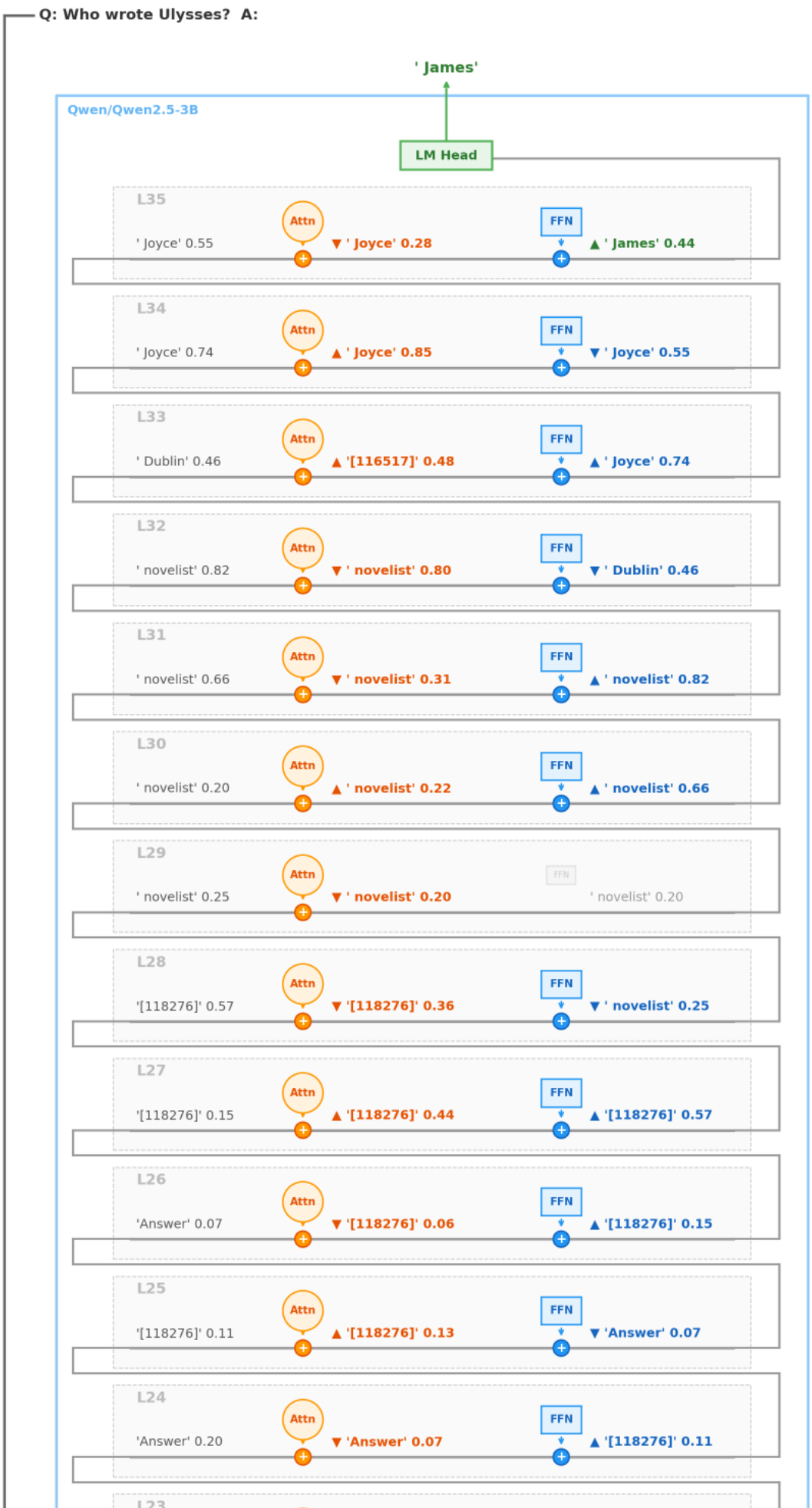
L30 — still "(answer)" but something is building. Attention starts pulling in information from the "Ulysses" token in the prompt.

L31–L32 — FFN delivers: **"James"** emerges as top-1. This is the first token of "James Joyce". Prob starts modest (~0.15) but climbs fast through L32–L33.

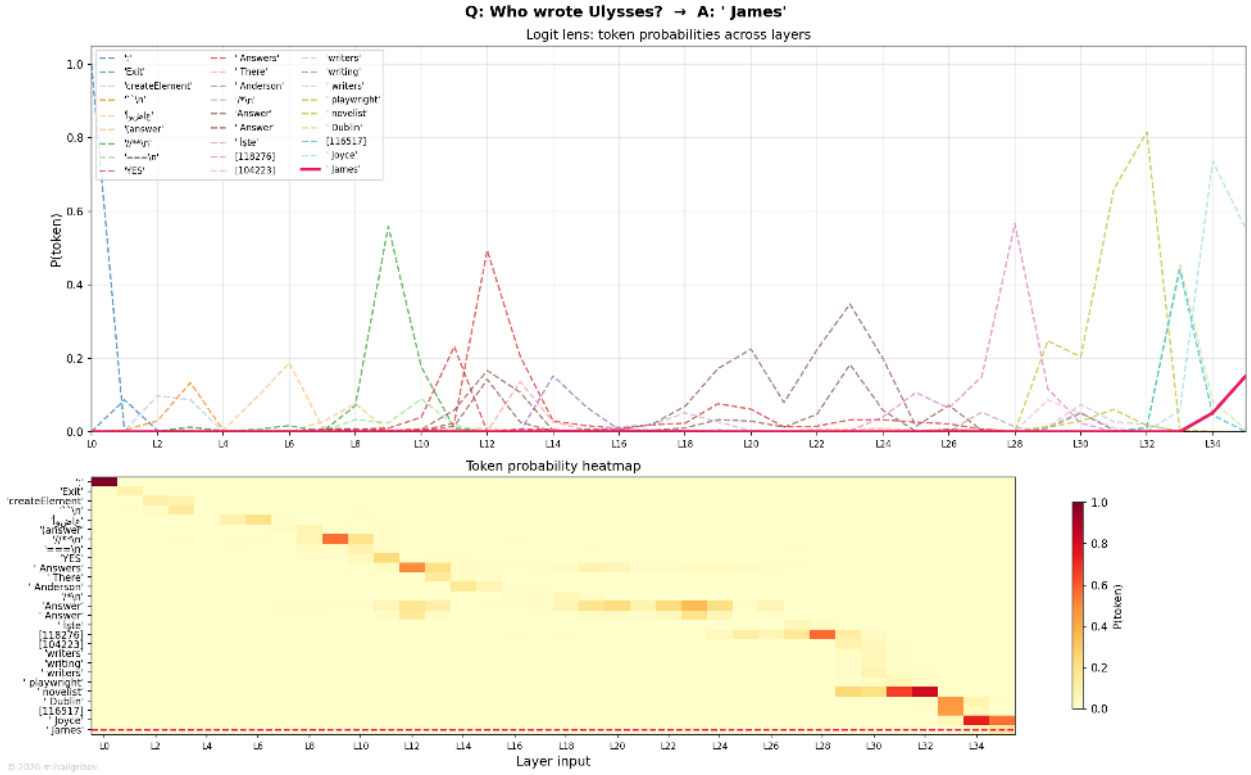
L33–L34 — rapid ascent. "James" prob reaches ~0.90. Both attention and FFN reinforce the answer. The model is now confident.

L35 — "James" holds with slight prob decrease (normal last-layer behavior).

Final output via LM Head: **James** (followed by " Joyce" in next generation step).


Q: Who wrote Ulysses? A:
' James'
Qwen/Qwen2.5-3B
LM Head
L35
' Joyce' 0.55
Attn
▼ ' Joyce' 0.28
FFN
▲ ' James' 0.44
L34
' Joyce' 0.74
Attn
▲ ' Joyce' 0.85
FFN
▼ ' Joyce' 0.55
L33
' Dublin' 0.46
Attn
▲ '[116517]' 0.48
FFN
▲ ' Joyce' 0.74
L32
' novelist' 0.82
Attn
▼ ' novelist' 0.80
FFN
▼ ' Dublin' 0.46
L31
' novelist' 0.66
Attn
▼ ' novelist' 0.31
FFN
▲ ' novelist' 0.82
L30
' novelist' 0.20
Attn
▲ ' novelist' 0.22
FFN
▲ ' novelist' 0.66
L29
' novelist' 0.25
Attn
▼ ' novelist' 0.20
FFN
' novelist' 0.20
L28
'[118276]' 0.57
Attn
▼ '[118276]' 0.36
FFN
▼ ' novelist' 0.25
L27
'[118276]' 0.15
Attn
▲ '[118276]' 0.44
FFN
▲ '[118276]' 0.57
L26
'Answer' 0.07
Attn
▼ '[118276]' 0.06
FFN
▲ '[118276]' 0.15
L25
'[118276]' 0.11
Attn
▲ '[118276]' 0.13
FFN
▼ 'Answer' 0.07
L24
'Answer' 0.20
Attn
▼ 'Answer' 0.07
FFN
▲ '[118276]' 0.11
L23

The chart reveals the literary competition — "writing", "playwright", "novelist" all flicker in mid-layers, showing the model exploring the author/literature domain. "James" (red) appears late at L31 and takes over decisively. Note that "Joyce" never appears because the model commits to the first name token first.



Which planet is closest to the Sun?

Token flow

L0–L10 — noise. No semantic content yet.

L11–L12 — Q&A format recognized. "Answers" (prob 0.50) then "Answer" takes over. Standard structural phase.

L13–L17 — astronomy domain emerges. Tokens like "planet", "Earth" briefly appear. The model knows it's about space but hasn't picked a planet yet. The competition between domain tokens is visible — "solar", "Earth", "planet" flicker without any gaining traction.

L18–L27 — "Answer" holds. Long accumulation phase. The residual stream is integrating information from "closest", "Sun", "planet" through attention across these layers. Prob fluctuates (0.05–0.25).

L28–L29 — transition zone. "(answer)" appears. The model is preparing for factual retrieval.

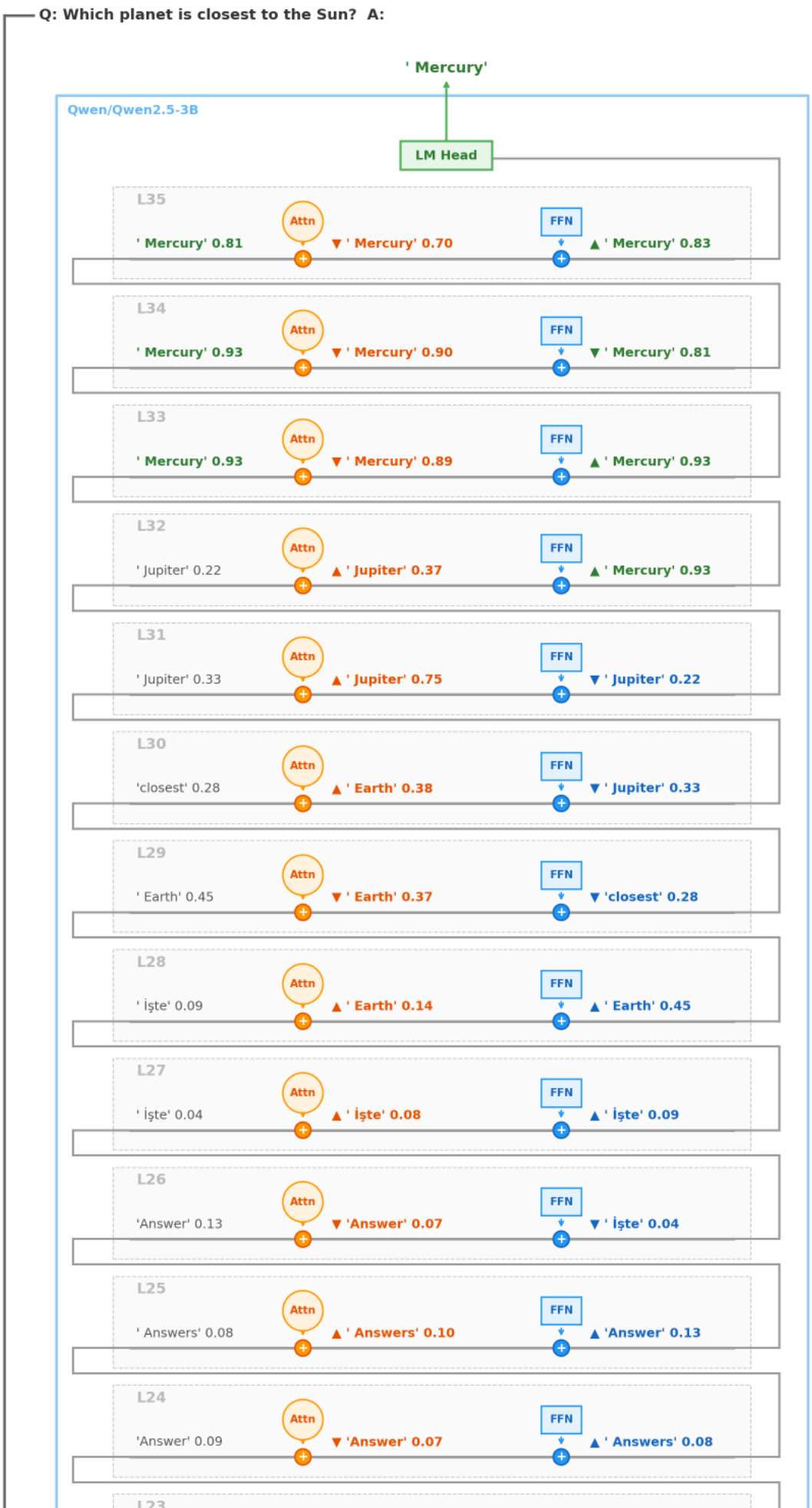
L30 — FFN fires: **"Mercury"** appears as top-1. This is the knowledge retrieval moment. The prob starts at ~0.30 — the model has found the answer but isn't fully confident yet.

L31–L32 — rapid climb. Mercury prob rises to 0.85, then 0.95. Both attention and FFN reinforce. The model is locking in.

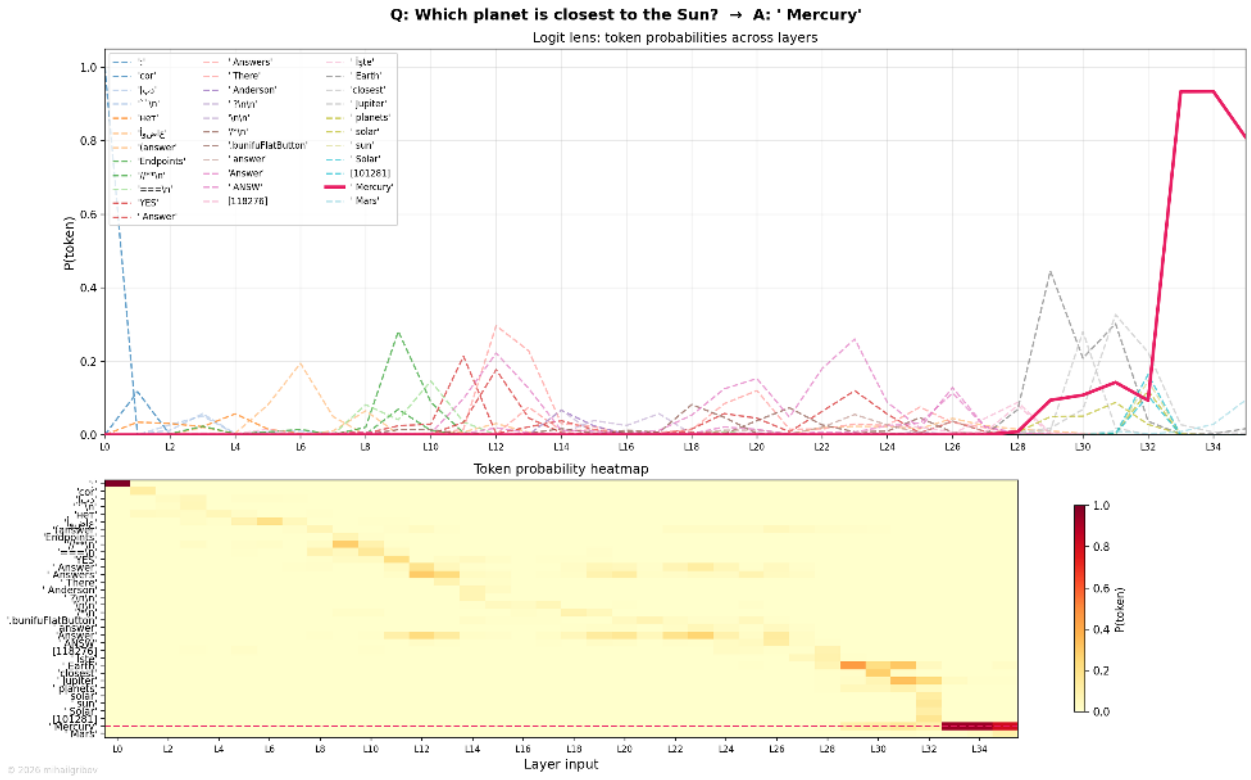
L33–L34 — near certainty. Prob ~0.99. The answer is stable.

L35 — slight prob decrease (typical last-layer formatting adjustment). Mercury holds.

Final output via LM Head: **Mercury**.


Q: Which planet is closest to the Sun? A:
' Mercury'
Qwen/Qwen2.5-3B
LM Head
L35
' Mercury' 0.81
Attn
▼ ' Mercury' 0.70
FFN
▲ ' Mercury' 0.83
L34
' Mercury' 0.93
Attn
▼ ' Mercury' 0.90
FFN
▼ ' Mercury' 0.81
L33
' Mercury' 0.93
Attn
▼ ' Mercury' 0.89
FFN
▲ ' Mercury' 0.93
L32
' Jupiter' 0.22
Attn
▲ ' Jupiter' 0.37
FFN
▲ ' Mercury' 0.93
L31
' Jupiter' 0.33
Attn
▲ ' Jupiter' 0.75
FFN
▼ ' Jupiter' 0.22
L30
'closest' 0.28
Attn
▲ ' Earth' 0.38
FFN
▼ ' Jupiter' 0.33
L29
' Earth' 0.45
Attn
▼ ' Earth' 0.37
FFN
▼ 'closest' 0.28
L28
' İşte' 0.09
Attn
▲ ' Earth' 0.14
FFN
▲ ' Earth' 0.45
L27
' İşte' 0.04
Attn
▲ ' İşte' 0.08
FFN
▲ ' İşte' 0.09
L26
'Answer' 0.13
Attn
▼ 'Answer' 0.07
FFN
▼ ' İşte' 0.04
L25
' Answers' 0.08
Attn
▲ ' Answers' 0.10
FFN
▲ 'Answer' 0.13
L24
'Answer' 0.09
Attn
▼ 'Answer' 0.07
FFN
▲ ' Answers' 0.08
L23

The chart shows many space-related tokens competing in mid-layers — "Earth", "planet", "solar" — none building sustained probability. Mercury (red) appears around L30 and climbs steeply. Clean factual retrieval with no competing answer in the late layers.



What is the lightest element?

Token flow

L0–L10 — noise. The residual stream carries random tokens with no relation to chemistry. This is normal — the model hasn't started processing the question semantically yet.

L11–L12 — the model recognizes the Q&A format. "Answers" (prob 0.50) dominates — this is a structural token, not an answer. The model is saying "I know I need to answer something".

L13–L14 — first interesting signal: **"helium"** briefly appears. The model is already in the right domain (elements) but picked the wrong one. This is the category-level retrieval — "something from periodic table".

L15–L27 — "Answer" takes over and holds for 12 layers. Nothing seems to happen, but internally the residual stream is accumulating information. The prob fluctuates (0.04–0.29) showing that the model is still searching.

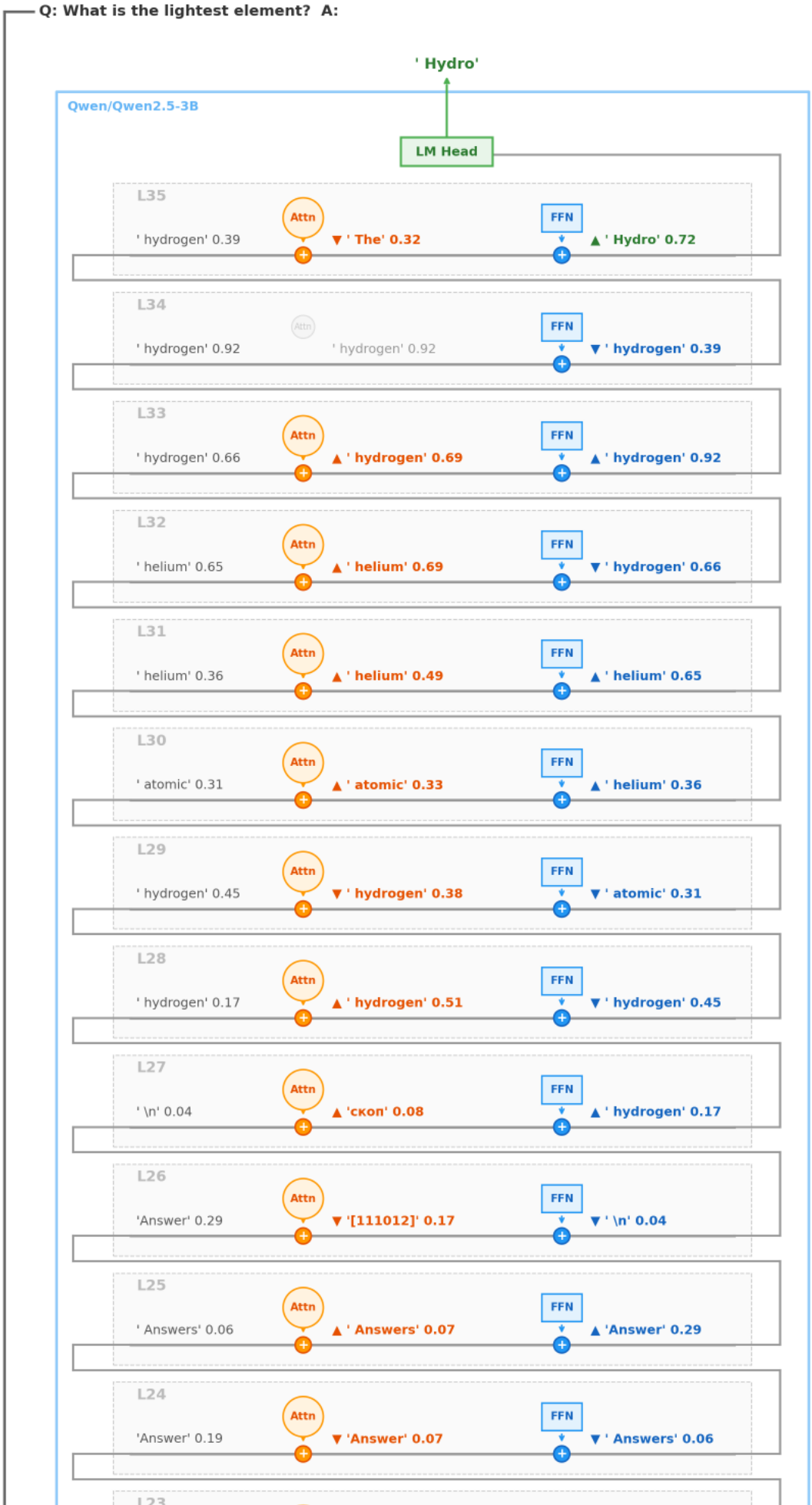
L28–L29 — the competition heats up. "(answer)" replaces "Answer" — a subtle shift suggesting the model is getting closer to committing to a specific response.

L30–L32 — FFN makes the call: **"hydrogen"** emerges at L30 and climbs fast. By L32, prob reaches 0.65. By L33, it's at 0.92. This is the knowledge retrieval moment — FFN neurons encoding "lightest + element → hydrogen" fire and push the answer through.

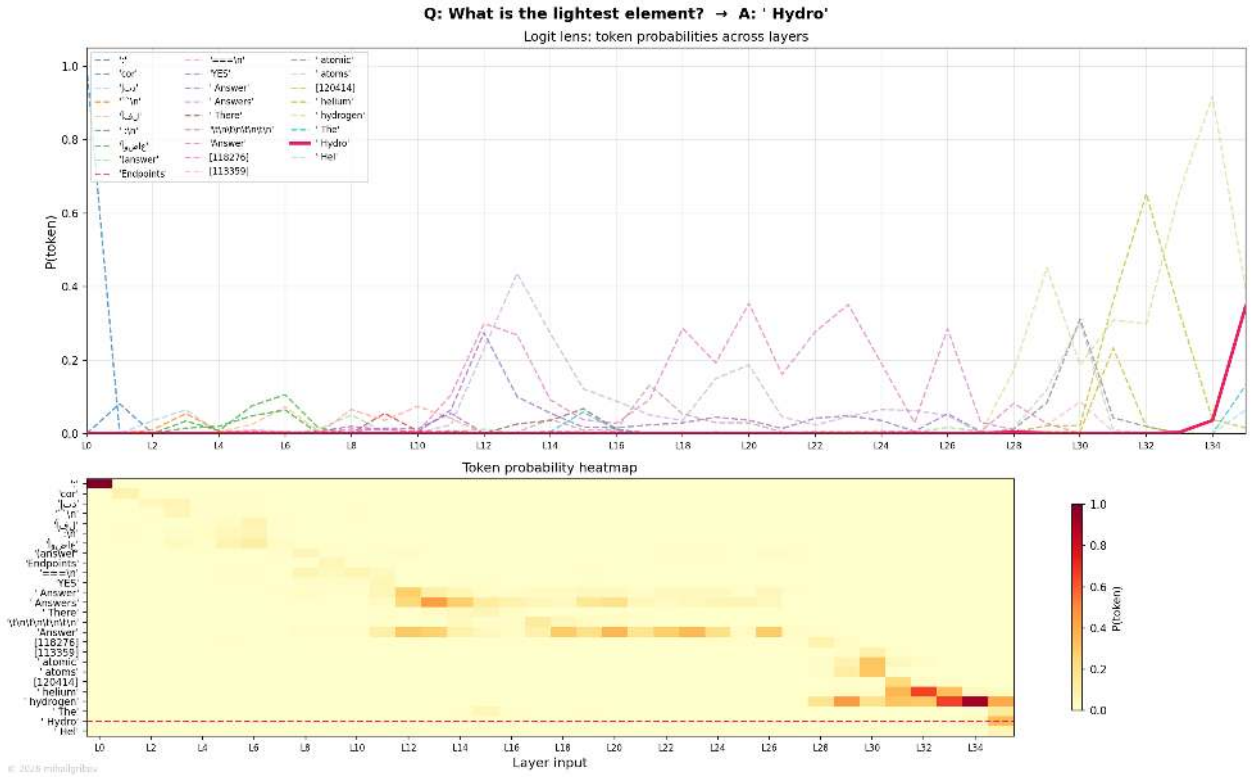
L33–L34 — stabilization. Both attention and FFN maintain hydrogen with minor adjustments. The answer looks settled.

L35 — a surprising twist. The model switches from "hydrogen" to **"Hydro"** — a different subword token that will be followed by "gen" to produce capitalized "Hydrogen". This isn't a change of answer — it's a tokenization decision. The model is choosing how to format the output ("Hydrogen" vs "hydrogen").

Final output via LM Head: **Hydro**(gen) → "Hydrogen".


Q: What is the lightest element? A:
' Hydro'
Qwen/Qwen2.5-3B
LM Head
L35
' hydrogen' 0.39
Attn
▼ ' The' 0.32
FFN
▲ ' Hydro' 0.72
L34
' hydrogen' 0.92
Attn
' hydrogen' 0.92
FFN
▼ ' hydrogen' 0.39
L33
' hydrogen' 0.66
Attn
▲ ' hydrogen' 0.69
FFN
▲ ' hydrogen' 0.92
L32
' helium' 0.65
Attn
▲ ' helium' 0.69
FFN
▼ ' hydrogen' 0.66
L31
' helium' 0.36
Attn
▲ ' helium' 0.49
FFN
▲ ' helium' 0.65
L30
' atomic' 0.31
Attn
▲ ' atomic' 0.33
FFN
▲ ' helium' 0.36
L29
' hydrogen' 0.45
Attn
▼ ' hydrogen' 0.38
FFN
▼ ' atomic' 0.31
L28
' hydrogen' 0.17
Attn
▲ ' hydrogen' 0.51
FFN
▼ ' hydrogen' 0.45
L27
' \n' 0.04
Attn
▲ 'скон' 0.08
FFN
▲ ' hydrogen' 0.17
L26
'Answer' 0.29
Attn
▼ '[111012]' 0.17
FFN
▼ ' \n' 0.04
L25
' Answers' 0.06
Attn
▲ ' Answers' 0.07
FFN
▲ 'Answer' 0.29
L24
'Answer' 0.19
Attn
▼ 'Answer' 0.07
FFN
▼ ' Answers' 0.06

The prob chart shows the full competition. Multiple candidates flicker in early/mid layers — none builds sustained probability. "hydrogen" (yellow-green dashed) appears around L30 and climbs steadily. At L35, "Hydro" (solid red) takes over — same answer, different tokenization path.



Who painted the Mona Lisa?

Token flow

L0–L10 — noise. Random tokens, no semantic processing.

L11–L12 — Q&A format recognized. "Answers" then "Answer". The model knows it needs to respond to a question.

L13–L17 — art domain activates. "painting", "painter", "artist" tokens appear and compete. The model has figured out the category — this is about visual art — but hasn't identified the specific artist. This is the category-level retrieval phase we see across all examples.

L18–L27 — "Answer" takes over for a long stretch. The residual stream is working internally — attention is pulling in context from "painted", "Mona Lisa" in the prompt, building the representation that will trigger the right knowledge neurons.

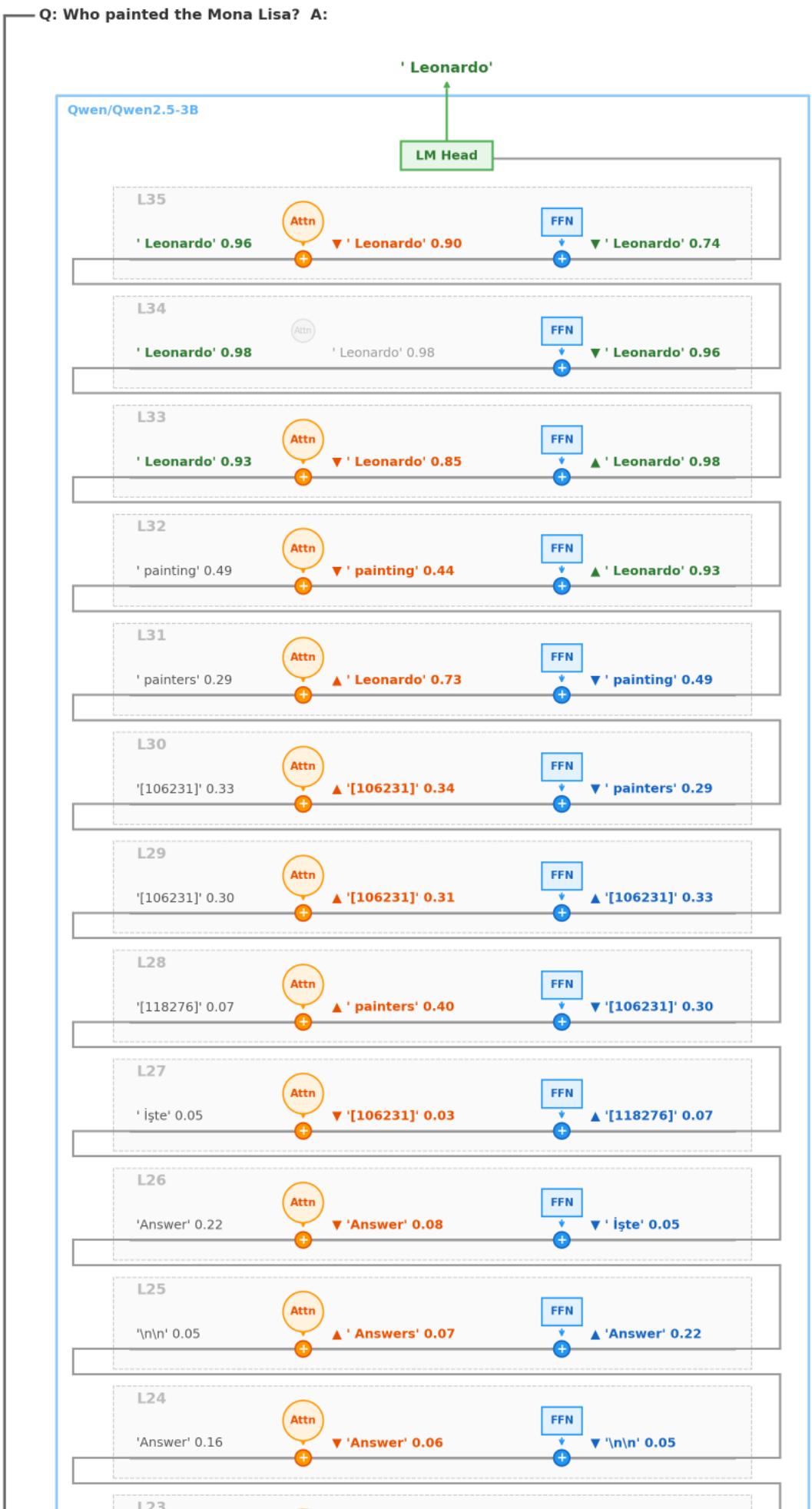
L28–L30 — transitional layers. Domain tokens reappear briefly — "painting" flickers. The competition between art-related tokens intensifies as the model prepares for final retrieval.

L31–L32 — FFN retrieves the answer: **"Leonardo"** emerges as top-1. The prob starts building — this is the first token of "Leonardo da Vinci". The FFN neurons encoding "Mona Lisa → Leonardo" have fired.

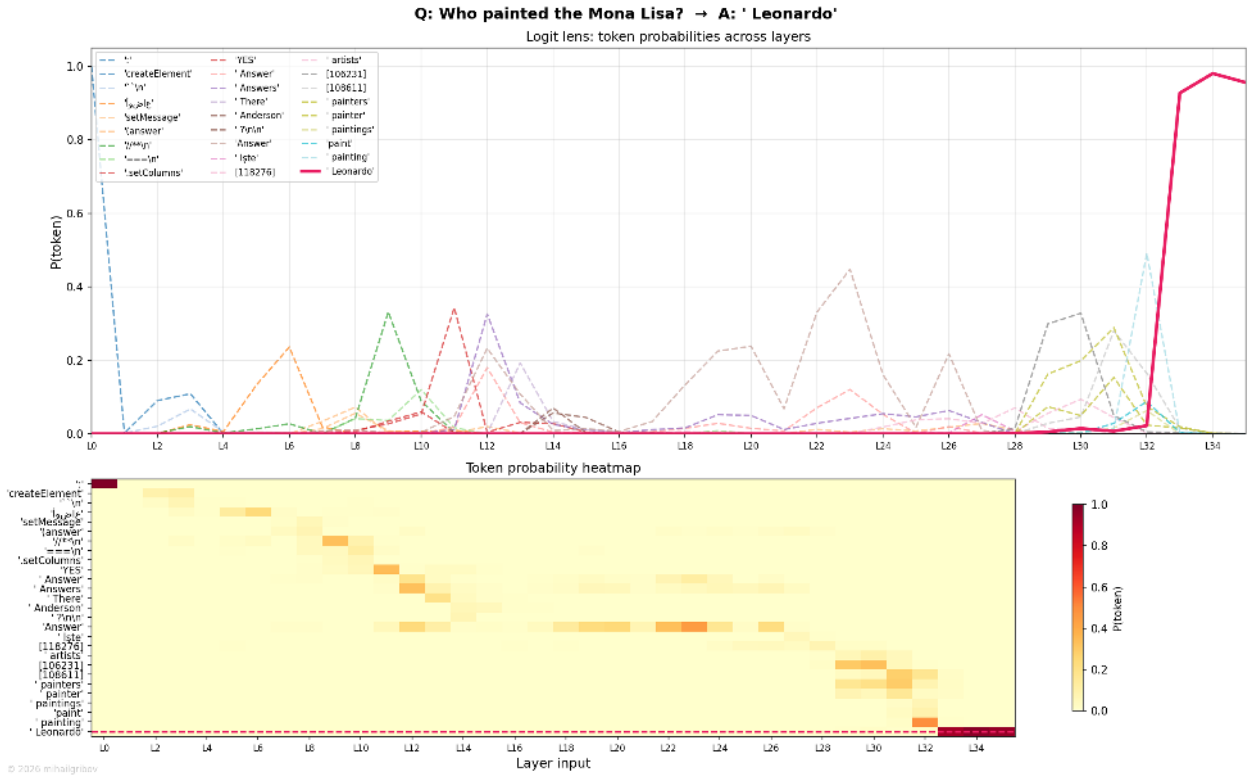
L33–L34 — rapid climb. "Leonardo" prob reaches ~0.90+. Both attention and FFN reinforce. The model is confident.

L35 — "Leonardo" holds through the final layer with minor prob adjustment.

Final output via LM Head: **Leonardo** (followed by " da Vinci" in subsequent generation steps).


Q: Who painted the Mona Lisa? A:
' Leonardo'
Qwen/Qwen2.5-3B
LM Head
L35
' Leonardo' 0.96
Attn
▼ ' Leonardo' 0.90
FFN
▼ ' Leonardo' 0.74
L34
' Leonardo' 0.98
Attn
' Leonardo' 0.98
FFN
▼ ' Leonardo' 0.96
L33
' Leonardo' 0.93
Attn
▼ ' Leonardo' 0.85
FFN
▲ ' Leonardo' 0.98
L32
' painting' 0.49
Attn
▼ ' painting' 0.44
FFN
▲ ' Leonardo' 0.93
L31
' painters' 0.29
Attn
▲ ' Leonardo' 0.73
FFN
▼ ' painting' 0.49
L30
'[106231]' 0.33
Attn
▲ '[106231]' 0.34
FFN
▼ ' painters' 0.29
L29
'[106231]' 0.30
Attn
▲ '[106231]' 0.31
FFN
▲ '[106231]' 0.33
L28
'[118276]' 0.07
Attn
▲ ' painters' 0.40
FFN
▼ '[106231]' 0.30
L27
' İşte' 0.05
Attn
▼ '[106231]' 0.03
FFN
▲ '[118276]' 0.07
L26
'Answer' 0.22
Attn
▼ 'Answer' 0.08
FFN
▼ ' İşte' 0.05
L25
'\n\n' 0.05
Attn
▲ ' Answers' 0.07
FFN
▲ 'Answer' 0.22
L24
'Answer' 0.16
Attn
▼ 'Answer' 0.06
FFN
▼ '\n\n' 0.05
L23

The chart shows a clear pattern: art-domain tokens ("painting", "painter") compete in mid-layers, representing the category search. "Leonardo" (red) doesn't appear until L31 but then rises decisively — the transition from "I know it's a painter" to "I know it's Leonardo" happens in just 2 layers.



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Source: github.com/mihail-gribov/llm_token_flows

Contact: mihail.gribov.rs@gmail.com · LinkedIn